Jacopo Zacchigna

MSc Student · Mechanistic Interpretability of LLMs · Trieste, IT

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Research Experience

Supervised Program for Alignment Research (SPAR), *Research Fellow* — Feb 2026 – May 2026

Research fellowship under *Gabriele Sarti* studying persona representations in LLMs (implicit-personalization).

- Extracted and released persona vectors for Gemma-2-27B and Llama-3.1-405B, enabling probing and steering of learned user representations.
- Developed `persona-data`, a Python package with dataset loaders and prompt utilities for the SynthPersona dataset (1k personas).
- Built an interactive web UI for real-time extraction, analysis, and probing of implicit persona representations.

Mechanistic Interpretability of Mixture-of-Experts LLMs, *Independent research* — Jan 2026 – present

- Adapting HeadPursuit / SOMP to identify expert specialization (e.g., toxicity) in OLMoE and GPT-OSS.
- Investigating expert editing and steering interventions versus knockout-style ablations.

Selected Projects

Causal Multi-Head Self-Attention Kernels, code — Jan – Feb 2026

- Implemented causal MHSA in CUDA, OpenMP, and SIMD on Leonardo (Cineca), iterating six CUDA kernels up to an online-softmax (FlashAttention-style) version.
- Scaled the OpenMP kernel to 128 cores with 94× speedup over its single-threaded baseline (73% parallel efficiency), 19× faster than naive PyTorch.

BayesianFlow Uncertainty Estimation in Generative Models, code — Jun – Jul 2025

- Extended BayesDiff to flow matching, achieving **5× faster** generation than DDIM at comparable quality.
- Integrated Last-Layer Laplace Approximation for pixel-wise uncertainty (*interpretable* confidence maps).

Publications

Zacchigna, J. & Liu, W. (co-first authors), et al. *Application of Computer Vision to the Automated Extraction of Metadata from Natural History Specimen Labels: A Case Study on Herbarium Specimens.* Plants (MDPI) — 2026

Education

University of Trieste, MSc, Data Science & Artificial Intelligence – Trieste, Italy — 2024 – 2026
Relevant coursework: Theory of Deep Neural Networks, Explainable AI, Probabilistic ML, Advanced Deep Learning.

University of Trieste, BSc, AI & Data Analytics – Trieste, Italy — 2021 – 2024

Experience

Obloo, AI Consultant – Trieste, Italy — May 2024 – Jan 2025

- Designed and deployed retrieval-augmented generation (RAG) systems over the company data lake.
- Mentored 5 interns to independent project delivery; led ML prototypes adopted in two company-wide products.

Skills

Technical: Python, C/C++, CUDA, PyTorch, nnsight/nnterp, Hugging Face, NumPy, R, Docker, Linux, HPC
Languages: Italian (native), English (C1)